TechNexa Solutions — Full App Test Report

Test date: 26 Oct 2026 **Environment:** AWS EC2 (k3s), live at `https://54.161.79.42.nip.io/` **Deployed commit:** `9efb808` (Jenkins build 51, SUCCESS) **Tested by:** AI QA agent (automated HTTP + browser checks)

> Roman Urdu summary: Puri app ka test kiya gaya — public pages, student/teacher portal, admin panel, chat/calls, enrollment, security aur SEO. Sab main routes 200 aa rahe hain. Koi blocking bug nahi mila. Ek minor item: HTTP->HTTPS global redirect optional hai (abhi HTTP 200). Call audio ke liye app-side fix deploy ho chuka hai; teacher ke device side par mic check karna zaroori hai.

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1. Public site

Check	Result
Home `/`	200
Courses `/courses.html`	200
About `/about.html`	200
Industries `/industries.html`	200
Legal `/legal.html`	200
Service pages (6)	200
404 page (unknown path)	200 branded 404
`robots.txt`	200, disallows `/portal/`, `/admin/`, `/api/`, `/uploads/`
`sitemap.xml`	200, no portal/admin URLs
`favicon.svg`	200
`/healthz`	200
HTTPS (nip.io)	Trusted cert, 200
HTTP (no redirect)	200 (global redirect optional, per design)

Notes: Contact form lives on the home page (no separate `contact.html` — expected). Enrollment form requires CNIC/photo/payment uploads.

2. SEO

- Canonical, OG, Twitter, JSON-LD present on public pages (per earlier audit).
- Portal and admin pages carry `noindex,nofollow` (verified on `/portal/messages` and `/admin/calls`).

- Internal link crawl: clean.

3. Student portal

All routes returned 200 for `umair799284@gmail.com`:

`/portal/dashboard`, `/portal/enrollments`, `/portal/assignments`, `/portal/quizzes`, `/portal/timetable`, `/portal/attendance`, `/portal/materials`, `/portal/documents`, `/portal/fees`, `/portal/profile`, `/portal/messages`.

- Login open-redirect fixed (verified: malicious `next` falls back to dashboard).
- Logout works.
- Dashboard shows enrollments, attendance, assignments, quizzes, fees.

4. Teacher portal

All routes returned 200 for `ibrahim@gmail.com`:

`/portal/dashboard`, `/portal/assignments`, `/portal/quizzes`, `/portal/timetable`, `/portal/attendance`, `/portal/materials`, `/portal/messages`, `/portal/profile`.

5. Admin panel

All routes returned 200 for `admin@cybernexa.local`:

`/admin/dashboard`, `/admin/courses`, `/admin/students`, `/admin/teachers`, `/admin/enrollments`, `/admin/assignments`, `/admin/campus`, `/admin/logs`, `/admin/calls`, `/admin/messages`.

- Admin dashboard attention queues present.
- Login activity page + CSV export working.
- Call logs page + CSV export working (13 calls today, deduplicated).

6. Chat & calls

- Realtime WebSocket connects over HTTPS/WSS.
- Messages send/receive, read ticks, typing indicator.
- Attachments: photo/document upload, in-app viewer, download affordances removed, right-click/drag blocked, audio `controlsList="nodownload"`.

- Attachment access: participant-only (non-participant 403, deleted 404) — verified.
- Delete-for-everyone removes file + DB row.
- Voice recording button present.
- Call log: caller/callee, type, status, duration, PKT time; filters + CSV.
- Call audio fix deployed:
    - Remote audio/video tracks now collected in one `MediaStream` (fixes one-way audio where video arrival dropped audio).
    - In-call microphone panel with live level + device switch.
    - Auto-switch from a silent saved mic to system default.
    - Yellow banners for muted mic / no sound from other person.
- Call lifecycle verified: answered (51s), rejected, missed (offline), WS-disconnect finalization.

Open item (device-side): If the teacher's mic is physically muted / OS-blocked / wrong device, the app now shows it in the mic panel. The teacher must act on it.

7. Enrollment flow

- Guest enrollment submission: validated (CNIC, photo, payment proof required).
- Duplicate pending request blocked (verified: second submission returns `duplicate: true`).
- Guest enrollment linked to new account by email (per earlier work).
- Admin enrollments page shows requests.

8. Forgot password

- `/portal/forgot` accepts email and redirects to `sent=1`.
- SMTP not configured → admin-mediated flow + WhatsApp fallback (per design).
- Admin reset queue + token copy available.

9. Security & privacy

- Login open-redirect fixed (malicious `next` falls back to dashboard).
- Portal/admin pages `noindex`.
- `robots.txt` blocks `/portal/`, `/admin/`, `/api/`, `/uploads/`.
- Chat attachments: server-side participant-only (403 for others); UI download removed; images/PDF open in private in-app viewer; audio `nodownload`.
- Delete-for-everyone purges file + DB.
- Call log updates restricted to call participants.

10. Bugs found & fixed during this QA

1. Call log duplicate rows from ICE-restart re-offers → deduplicated + unique index on `call_key`.
2. Call participants could be swapped on re-offer → map now preserved from first offer / DB.
3. One-way call audio (remote audio dropped when video arrived) → fixed via track accumulation.
4. Call `ended_at` not set when callee offline → fixed.
5. WS disconnect didn't finalize call logs → fixed.
6. `answered_at` was not recorded → added column + migration.

11. Items not blocking but recommended

1. Global HTTP→HTTPS redirect (optional, after final domain).
2. Real Facebook/Instagram/LinkedIn URLs in footer (placeholders remain `#`).
3. DuckDNS `webnexa.duckdns.org` A record → EC2 IP (for branded HTTPS).
4. SMTP credentials in secret `cybernexa-env` for self-service password reset emails.
5. Office files (`.docx` / `.xlsx`) still open natively in the browser (browser limitation); restrict them in chat if stricter privacy is wanted.
6. Chat text copy/select is still allowed; disable if needed.

12. Cleanup performed

- QA temp student (`qa.privacy.20261026@example.com`) deleted.
- QA call test rows/users (`qa_dup_*`) deleted.
- Enrollment test row deleted.
- No QA artifacts left in DB.
- Original chat intact (60 messages, 7 attachments).

13. Conclusion

No blocking defects found. All main routes (public, student, teacher, admin) return 200. Chat, calls, enrollment, and security flows behave as designed. The remaining items are configuration/ optional enhancements, not bugs.

Overall status: PASS